EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

The Window Method is the classical analytical approach to FIR filter design. Starting from an ideal brick-wall frequency response $H_d(e^{j\omega})$, its infinite non-causal impulse response $h_d[n]$ is truncated and smoothed using a finite window function $w[n]$.

Pedagogical Strategy:

1. Derive the ideal lowpass impulse response: $h_d[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$ where $\tau = \frac{N-1}{2}$.
2. Explain the **Gibbs Phenomenon**: Truncating with a rectangular window causes 8.95% peak overshoot at band edges, which does not vanish even as $N \to \infty$.
3. Analyze the fundamental trade-off: **Mainlobe width (transition bandwidth) vs. Sidelobe level (stopband attenuation)**.
4. Define and analyze Rectangular, Bartlett (triangular), and Hann (Hanning) windows.
5. Complete step-by-step filter design numericals.

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** ideal impulse responses $h_d[n]$ for Lowpass, Highpass, Bandpass, and Bandstop filters.
2. **Explain** the origins and mathematical properties of the Gibbs phenomenon.
3. **Design** linear-phase FIR filters using Rectangular, Bartlett, and Hann windows to meet transition bandwidth and attenuation specs.
4. **Evaluate** the frequency spectrum of window functions.

2. MATHEMATICAL FOUNDATIONS

2.1 Ideal Filter Impulse Responses ($\tau = \frac{N-1}{2}$)

- **Lowpass (ω_c):**

$$h_{d,\mathrm{LP}}[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$$

- **Highpass (ω_c):**

$$h_{d,\text{HP}}[n] = \delta[n-\tau] - \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$$

- **Bandpass (ω_{c1}, ω_{c2}):**

$$h_{d,\text{BP}}[n] = \frac{\sin(\omega_{c2}(n-\tau))}{\pi(n-\tau)} - \frac{\sin(\omega_{c1}(n-\tau))}{\pi(n-\tau)}$$

2.2 Window Functions Formulation ($0 \le n \le N-1$)

1. **Rectangular Window:**

$$w[n] = 1, \quad \Delta\omega = \frac{4\pi}{N}, \quad \text{Peak Sidelobe: } -13 \text{ dB}, \quad A_s = 21 \text{ dB}$$

2. **Bartlett (Triangular) Window:**

$$w[n] = 1 - \frac{2|n-\tau|}{N-1}, \quad \Delta\omega = \frac{8\pi}{N}, \quad \text{Peak Sidelobe: } -25 \text{ dB}, \quad A_s = 25 \text{ dB}$$

3. **Hann (Hanning) Window:**

$$w[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{N-1}\right), \quad \Delta\omega = \frac{8\pi}{N}, \quad \text{Peak Sidelobe: } -31 \text{ dB}, \quad A_s = 44 \text{ dB}$$

---

3. WORKED NUMERICAL EXAMPLES

Example 22.1: FIR Lowpass Filter Design using Hann Window

Problem: Design a causal FIR lowpass filter with cutoff $\omega_c = \frac{\pi}{4}$ rad/sample and length $N = 7$ using a Hann window.

Solution:

- Length $N = 7 \implies \tau = \frac{7-1}{2} = 3$.
- Ideal impulse response:

$$h_d[n] = \frac{\sin(\frac{\pi}{4}(n-3))}{\pi(n-3)}$$

  - $h_d[3] = \frac{\omega_c}{\pi} = \frac{\pi/4}{\pi} = 0.2500$
  - $h_d[2] = h_d[4] = \frac{\sin(-\pi/4)}{-\pi} = \frac{\sqrt{2}/2}{\pi} = \frac{0.7071}{\pi} = 0.2251$
  - $h_d[1] = h_d[5] = \frac{\sin(-2\pi/4)}{-2\pi} = \frac{1}{2\pi} = 0.1592$
  - $h_d[0] = h_d[6] = \frac{\sin(-3\pi/4)}{-3\pi} = \frac{0.7071}{3\pi} = 0.0750$
- **Hann Window Values ($w[n] = 0.5 - 0.5\cos(\frac{2\pi n}{6})$):**
  - $w[3] = 0.5 - 0.5\cos(\pi) = 1.0000$
  - $w[2] = w[4] = 0.5 - 0.5\cos(2\pi/3) = 0.5 - 0.5(-0.5) = 0.7500$
  - $w[1] = w[5] = 0.5 - 0.5\cos(\pi/3) = 0.5 - 0.5(0.5) = 0.2500$
  - $w[0] = w[6] = 0.5 - 0.5\cos(0) = 0.0000$
- **Final FIR Filter Coefficients ($h[n] = h_d[n] \cdot w[n]$):**
  - $h[0] = h[6] = 0.0750 \times 0.0000 = 0.0000$

- $h[1] = h[5] = 0.1592 \times 0.2500 = 0.0398$
- $h[2] = h[4] = 0.2251 \times 0.7500 = 0.1688$
- $h[3] = 0.2500 \times 1.0000 = 0.2500$

$$h[n] = \underset{\uparrow}{0.0000}, 0.0398, 0.1688, 0.2500, 0.1688, 0.0398, 0.0000$$

---

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) What is the Gibbs phenomenon? Explain how window functions reduce Gibbs ringing at the expense of transition bandwidth. *(6 Marks)* **(b)** Design a 9-tap FIR Highpass filter with cutoff $\omega_c = \frac{\pi}{3}$ using a Bartlett window. *(9 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Explanation of 8.95% overshoot at step discontinuities and Fourier truncation *(3 Marks)*
  - Tapering effect of windows: Smooth boundary transitions suppress high-frequency sidelobes but widen mainlobe transition band *(3 Marks)*
- **Part (b):**
  - $N = 9, \tau = 4.$
  - $h_d[n] = \delta[n-4] - \frac{\sin(\pi(n-4)/3)}{\pi(n-4)}$ *(3 Marks)*
  - Bartlett window: $w[n] = 1 - \frac{|n-4|}{4}$ for $n = 0, \ldots, 8$ *(2 Marks)*
  - Evaluation of all 9 coefficients $h[n] = h_d[n]w[n]$ with symmetry *(4 Marks)*

---

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

N = 7
tau = 3
wc = np.pi / 4
n = np.arange(N)

# Ideal LPF
hd = np.zeros(N)
hd[n ≠ tau] = np.sin(wc * (n[n ≠ tau] - tau)) / (np.pi * (n[n ≠ tau] - tau))
hd[tau] = wc / np.pi

# Hann window
w = 0.5 - 0.5 * np.cos(2 * np.pi * n / (N - 1))
h = hd * w

print("Designed FIR Filter Coefficients h[n]:")
print(np.round(h, 4))
```